

BAB_GJR_Report

Betting Against Beta with Conditional Volatility

A GJR-GARCH Refinement of Frazzini & Pedersen (2014)

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Period covered. January 2000 – December 2022

In-sample (IS). 2000–2015 — **Out-of-sample (OOS).** 2016–2022 (84 months)

Abstract

Frazzini and Pedersen (2014) show that low-beta stocks deliver risk-adjusted returns above what CAPM predicts, and that a leverage-neutral long-low / short-high portfolio (BAB) earns a robust premium. Their estimator is built on rolling-window OLS, which assumes the joint distribution of stock and market returns is approximately stable. We replace it with a conditional one. Market variance is filtered with a GJR-GARCH(1,1) under Student-t innovations and stock–market covariances with EWMA at $\lambda = 0.94$. Conditional betas are then $\beta_{i,T}^{cond} = \widehat{Cov}_{i,T} / \hat{\sigma}_{m,T}^2$. Out-of-sample (2016–2022), the GJR-based portfolio attains a gross annualised Sharpe ratio of **0.64** versus **0.19** for the OLS benchmark; net of 5 bp/side transaction costs, the figures fall to **0.55** and **0.17**. The OOS CAPM alpha rises from 16 bp/month ($t = 0.70$) for OLS to 51 bp/month ($t = 1.64$) for GJR, with the FF3 alpha showing the same pattern (9 bp vs 46 bp). Maximum drawdown over OOS contracts from -25.66% to -13.44% . The Jobson-Korkie (1981) test on the OOS Sharpe difference returns $z = 0.979$ ($p = 0.328$) — we cannot reject equality at conventional levels, a power issue with 84 monthly observations rather than evidence the gap is illusory. The cross-sectional correlation between the two beta estimates is only 0.334, and the GJR cross-section is materially wider (std 0.498 vs 0.314). The conditional estimator, in short, ranks stocks differently and the OOS evidence favours that ranking — at the cost of materially higher turnover.

1. Introduction and motivation

1.1 The low-beta anomaly

CAPM predicts a flat trade-off between beta and expected return: a stock with twice the beta should earn twice the market risk premium. Empirically, the line is too flat — high-beta stocks underperform their CAPM-implied return and low-beta stocks overperform. Black (1972) attributed this to leverage constraints: investors who want more equity exposure than their wealth allows must overweight high-beta stocks instead of borrowing, which pushes those stocks' prices up and their expected returns down. Frazzini and Pedersen (FP, 2014) formalised the mechanism and built a strategy — Betting Against Beta — that buys low-beta stocks levered up to a beta of one and shorts high-beta stocks de-levered to a beta of one. The portfolio is market-neutral by construction and earns a positive expected return whenever the security market line is too flat.

1.2 Where rolling OLS struggles

The FP estimator combines a one-year rolling volatility with a five-year rolling correlation, then shrinks toward one. This is parsimonious and robust on average, but two structural limitations matter for our test. First, the five-year correlation window smooths through regime changes: in late 2008 the OLS beta of a typical large-cap is still anchored to a covariance that is mostly pre-crisis. Second, the one-year volatility window weights all 252 trading days equally, so a calm month and a panic month enter symmetrically. Both effects bias the BAB ranking toward stale information when the cross-section of betas is changing fastest.

1.3 What conditional volatility brings

GJR-GARCH (Glosten, Jagannathan, and Runkle, 1993) lets variance respond asymmetrically to news: a large negative shock raises tomorrow's variance more than a positive shock of the same size. This is empirically the dominant feature of equity-index volatility — the leverage effect — and it concentrates exactly in the periods where the OLS estimator is slowest to react. Pairing a GJR market variance with an EWMA covariance (RiskMetrics, 1996) yields a beta that updates daily and therefore reflects the cross-sectional reordering of risk that occurs during stress. Our hypothesis is simple: a faster, asymmetry-aware estimator should produce a cleaner BAB ranking, and the improvement should concentrate in periods where market volatility is far from its long-run mean.

2. Data and sample

We use daily total returns from the CRSP Daily Stock File via WRDS, restricted to ordinary common shares (`shrcd` $\in$ {10, 11}) listed on NYSE, AMEX, or NASDAQ (`exchcd` $\in$ {1, 2, 3}). The investable universe is filtered each month-end to stocks with prior month-end market capitalisation above USD 200M. After the filter the panel contains 15.1 million stock-day observations covering 8,812 unique securities; the active cross-section varies from approximately 2,000 to 3,200 stocks per month (Figure A.1). The market proxy is the CRSP value-weighted index (`vwret` from `dsix`). The risk-free rate and Fama-French factors come from Kenneth French's data library. The sample runs January 2000 to December 2022 and is split into an in-sample (IS) period 2000–2015 used to estimate the GJR-GARCH parameters and to develop the strategy, and an out-of-sample (OOS) period 2016–2022 used purely for evaluation. No look-ahead crosses this boundary: the GJR parameters fitted on IS are frozen and applied recursively to OOS data.

A few cleaning steps deserve mention. We compute log-returns $r_{i,t} = \ln(1 + R_{i,t})$ and use them throughout volatility/covariance estimation. Survivorship bias is handled by allowing entries (IPOs) and exits (delistings) to flow through the universe — a stock enters the cross-section in month T if and only if its lagged market cap exceeds the threshold and it has at least 120 daily observations for volatility and 750 for correlation. The result is an unbalanced panel that mirrors what an investor would have actually traded. Because the OLS estimator requires five years of correlation history to fire, the first usable BAB observation is January 2003. The head-to-head comparison is therefore over 240 monthly observations (156 IS, 84 OOS).

3. Methodology

3.1 Benchmark estimator: rolling OLS beta

Following Frazzini and Pedersen (2014), at each month-end T we estimate the time-series beta as the product of the correlation and the volatility ratio,

$$\hat{\beta}_{i,T}^{TS} = \hat{\rho}_{i,T} \times \frac{\hat{\sigma}_{i,T}}{\hat{\sigma}_{m,T}}.$$

Rolling stock and market volatilities use a 252-day window with a 120-day minimum. The correlation $\hat{\rho}_{i,T}$ uses a 1,260-day (5-year) window of three-day overlapping log-returns with a 750-day minimum. The three-day overlap follows FP and reduces sensitivity to non-synchronous trading. The raw beta is shrunk toward one (Vasicek, 1973):

$$\hat{\beta}_{i,T} = 0.6 \times \hat{\beta}_{i,T}^{TS} + 0.4 \times 1.$$

Shrinkage toward unity is a Bayesian device: it penalises extreme beta estimates that are likely to contain noise. We apply the same shrinkage to the conditional estimator so that any difference in performance is

attributable to the conditional information set, not to the shrinkage. Final betas are clipped to $[0.1, 3.0]$ to prevent infinite leverage when an estimate degenerates.

3.2 Conditional estimator: GJR-GARCH market variance

Let $r_{m,t} = \mu_m + \varepsilon_{m,t}$ with $\varepsilon_{m,t} = \sigma_{m,t} z_t$, $z_t \sim t_\nu$, and

$$\sigma_{m,t}^2 = \omega + \alpha \varepsilon_{m,t-1}^2 + \gamma \varepsilon_{m,t-1}^2 \cdot \mathbf{1}_{\{\varepsilon_{m,t-1} < 0\}} + \beta \sigma_{m,t-1}^2.$$

The asymmetry parameter γ is the entire point. When yesterday's market shock was negative, tomorrow's variance loads on $\alpha + \gamma$ instead of α . Stationarity requires $\omega > 0$, $\alpha, \gamma, \beta \geq 0$ and $\alpha + \gamma/2 + \beta < 1$. Parameters are estimated once by maximum likelihood on IS data only (with the data scaled by 100 for numerical stability) and then frozen. Daily filtering on the full 2000–2022 period is a deterministic recursion initialised at the unconditional variance $\omega/(1 - \alpha - \gamma/2 - \beta)$. Student-t innovations are used because Gaussian residuals systematically underestimate the tails of equity-index returns.

3.3 Conditional covariance: EWMA

We estimate the daily covariance between each stock and the market with an exponentially weighted moving average,

$$\widehat{\text{Cov}}_{i,t} = \lambda \cdot \widehat{\text{Cov}}_{i,t-1} + (1 - \lambda) \cdot \varepsilon_{i,t} \cdot \varepsilon_{m,t},$$

with $\lambda = 0.94$ as in RiskMetrics (1996). The market residual is $\varepsilon_{m,t} = r_{m,t} - \mu_m$ using the GJR-fitted mean. The stock residual uses a zero-mean convention $\varepsilon_{i,t} = r_{i,t}$ which avoids look-ahead from full-sample means and is standard in this literature. The recursion is initialised on the first 125 trading days (~6 months) using the sample covariance.

3.4 Conditional beta and final shrinkage

At each month-end T ,

$$\beta_{i,T}^{\text{cond}} = \frac{\widehat{\text{Cov}}_{i,T}}{\hat{\sigma}_{m,T}^2}.$$

The same Vasicek shrinkage and the same $[0.1, 3.0]$ clip used for the OLS beta are then applied — by design, so that any difference in performance traces to the information set, not to differences in regularisation.

3.5 BAB portfolio construction

The construction is identical for both estimators. Within each cross-section at month-end T , we (i) rank stocks by beta; (ii) compute centred ranks $z_{i,T} = \text{rank}_{i,T}/(N + 1) - \text{median}$; (iii) form weights proportional to the positive part of the centred rank — long for stocks below the median, short for stocks above — normalised so each leg sums to one; (iv) compute the leg betas $\hat{\beta}_T^L = \sum w_{i,T}^L \hat{\beta}_{i,T}$ and $\hat{\beta}_T^H = \sum w_{i,T}^H \hat{\beta}_{i,T}$; and (v) build the BAB return

$$r_{T+1}^{\text{BAB}} = \frac{1}{\hat{\beta}_T^L} (r_{T+1}^L - r_f) - \frac{1}{\hat{\beta}_T^H} (r_{T+1}^H - r_f).$$

Leg betas are floored at $1/1.5 \approx 0.667$ so that no leg's leverage exceeds $1.5\times$. This cap matches what an institutional fund could plausibly hold and prevents pathological months where one leg's aggregated beta drops to near zero.

3.6 Transaction costs

We charge a per-side proportional fee of 5 basis points on monthly turnover:

$$TC_T = c \times \sum_i (|w_{i,T}^L - w_{i,T-1}^L| + |w_{i,T}^H - w_{i,T-1}^H|), \quad c = 0.0005.$$

Net returns are $r_T^{BAB,net} = r_T^{BAB} - TC_T$. This is conservative for institutional execution but realistic for retail benchmarks.

4. Empirical results

4.1 GJR-GARCH parameter estimates

Maximum-likelihood estimation on IS market returns produces the parameters in Table 1. The asymmetry coefficient $\gamma = 0.163$ is highly significant ($p < 0.0001$) — the canonical leverage effect. Two features stand out. First, the symmetric ARCH term α collapses to the zero boundary, meaning effectively all of the news impact in this sample is asymmetric: positive shocks barely raise next-day variance, negative shocks raise it by γ . Second, persistence sums to 0.984, very close to the unit-root boundary of an integrated-GARCH process and consistent with the long-memory behaviour of large-cap volatility. Filtered conditional volatility tracks 21-day realised volatility closely in calm regimes and rises asymmetrically through 2008, 2011, 2018 and 2020 (Figure 1). Across the full 2000–2022 period, the daily GJR volatility series has a mean of 1.09% and a maximum of 7.39%, equivalent to annualised values of approximately 17% and 117%.

Table 1 — GJR-GARCH(1,1)-t market model, IS estimates (data scaled by 100)

Parameter	Estimate	Notes
μ_m	0.0359	Daily mean ($\approx 9\%$ annualised)
ω	0.0168	Long-run variance anchor
α	0.0000	Symmetric ARCH (boundary)
γ	0.1630	Asymmetric ARCH ($p < 0.0001$)
β	0.9025	GARCH persistence
Persistence $\alpha + \gamma/2 + \beta$	0.9840	Stationary, near unit root

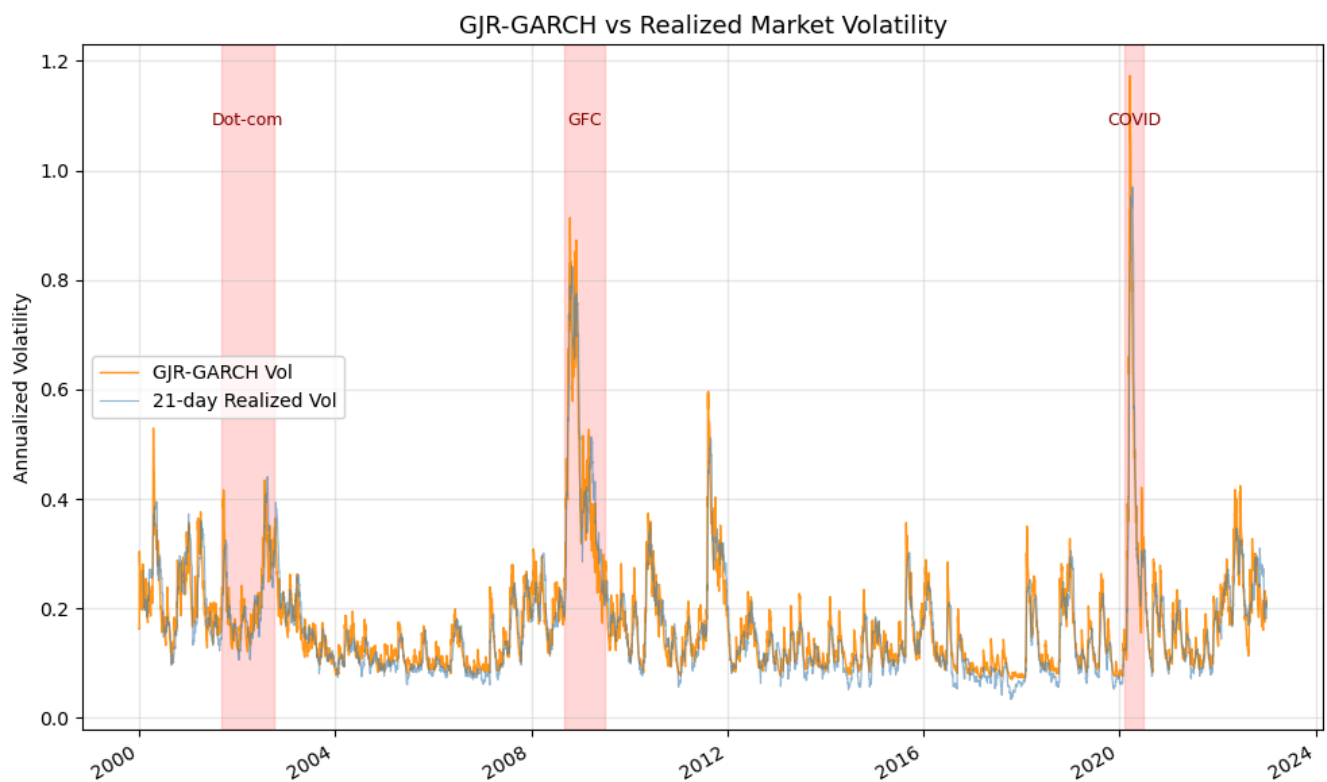


Figure 1: GJR-filtered annualised market volatility versus 21-day realised volatility, 2000–2022. Dot-com, GFC and COVID periods shaded. The two series track closely; GJR is a smoother, asymmetry-aware filter of the same underlying process.

4.2 Beta dynamics: OLS vs. conditional

The two estimators agree on the unconditional cross-sectional mean (1.123 for OLS, 1.113 for GJR) but disagree sharply on dispersion and ranking (Table 2). The GJR cross-section is materially wider — its standard deviation is 0.498 against 0.314 for OLS. More importantly, the cross-sectional correlation between the two beta series is only **0.334**: they rank stocks very differently. This is the first piece of evidence that the difference between the two BAB strategies is not a small perturbation of an otherwise common ranking, but two genuinely different views of the cross-section.

Table 2 — Cross-sectional beta statistics (pooled, all months)

	OLS $\hat{\beta}$	GJR β^{cond}
Count	484,326	678,078
Mean	1.123	1.113
Std	0.314	0.498
25% / 50% / 75%	0.91 / 1.08 / 1.28	0.78 / 1.02 / 1.34
Min / Max	0.10 / 3.00	0.10 / 3.00
Correlation (OLS, GJR)	\multicolumn{2}{c}{\mathbf{0.334}}	

The asymmetry between counts (GJR has more month-stock observations) reflects the fact that OLS requires five years of correlation history to fire, while GJR fires after the 125-day EWMA initialisation. Figure 2 shows the cross-sectional histograms at three reference dates: pre-GFC (June 2007), GFC peak (December 2008) and COVID (March 2020). The GJR distribution shifts and widens between calm and stressed periods; the OLS distribution barely moves because four years of pre-crisis data still dominate the five-year correlation window. Figure 3 plots OLS beta against GJR beta coloured by sub-period and confirms that the correlation between them is genuinely low, especially in the stressed years.

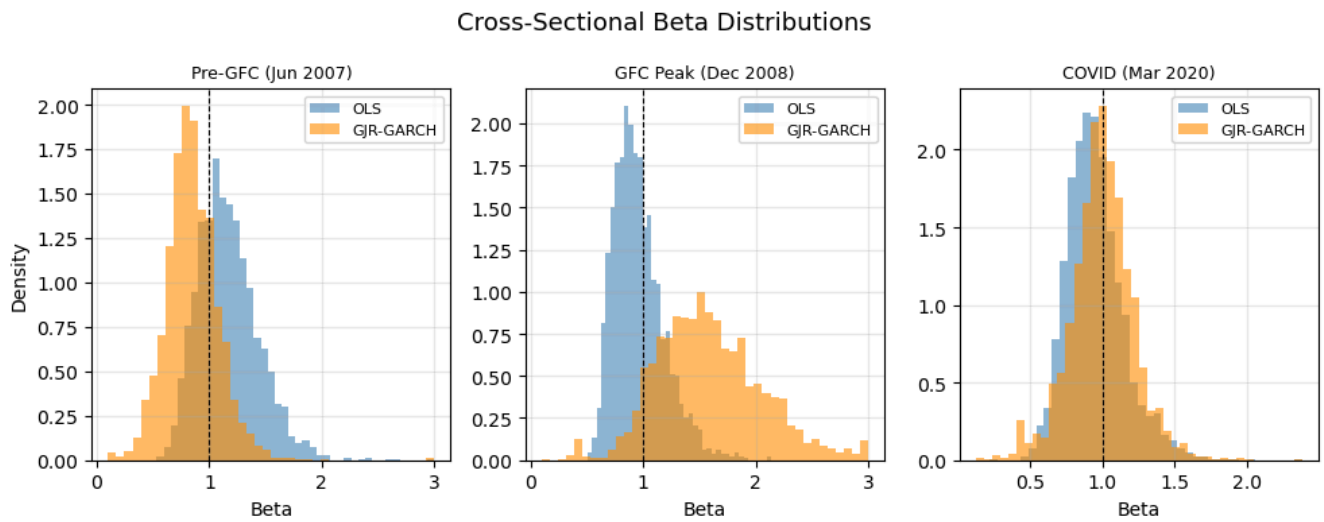


Figure 2: cross-sectional histograms of OLS vs. GJR conditional betas at three dates.

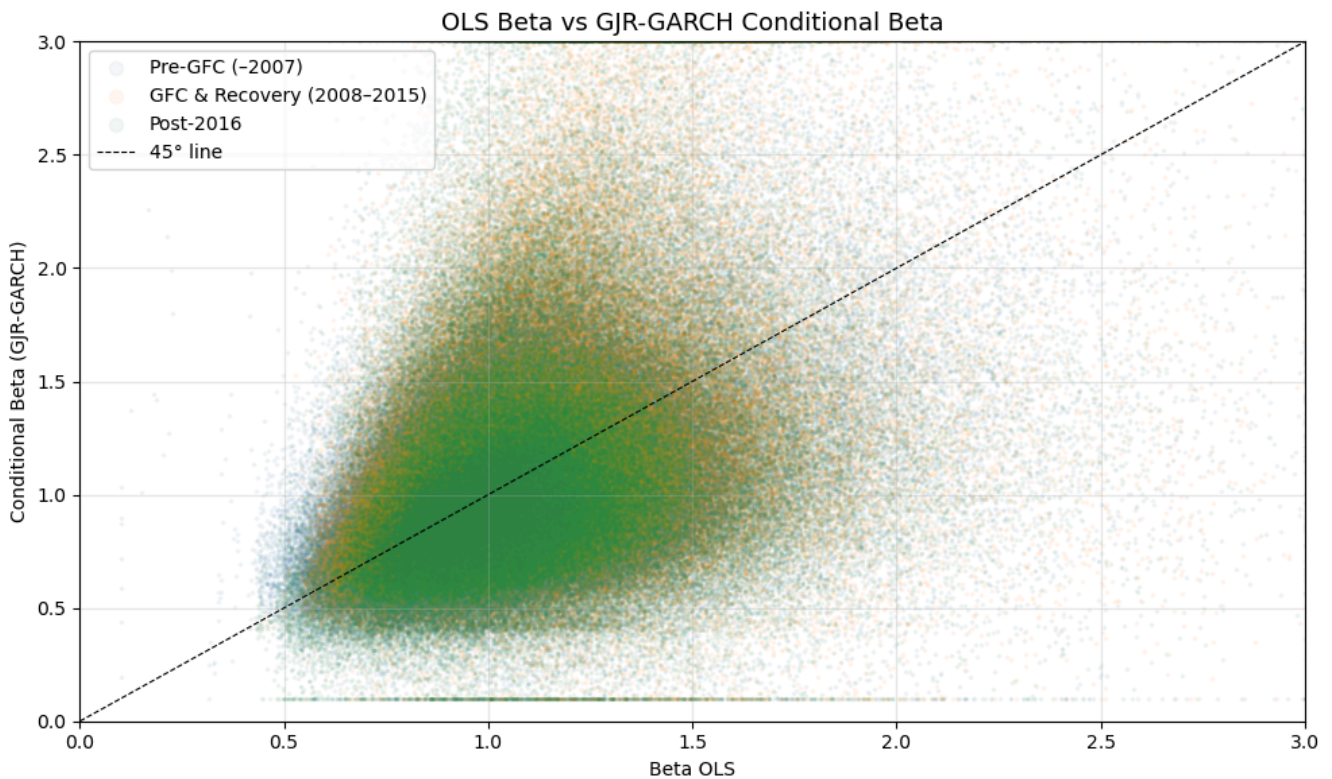


Figure 3: scatter of OLS beta against GJR conditional beta, coloured by sub-period (pre-GFC, GFC + recovery, post-2016). The cloud is diffuse across all regimes — the two estimators routinely disagree about which stocks are low- and which are high-beta.

4.3 Performance: full sample, IS and OOS

Table 3 reports the headline performance metrics. Over the 240-month common period (Jan 2003 – Dec 2022), BAB-GJR beats BAB-OLS in every horizon and on every metric except CVaR. The OOS gap is the largest: an annualised Sharpe of 0.64 against 0.19, a quadrupling of the annualised return (6.44% vs. 1.68%), and a maximum drawdown that contracts from –25.66% to –13.44%. The OOS skewness of BAB-GJR is mildly positive (+0.13) against –0.65 for OLS, an unusual feature for a cross-sectional equity strategy and a sign that the conditional estimator is not loading on stocks that crash en masse.

Table 3 — Performance summary, gross of transaction costs, 2003–2022

Strategy	Period	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	Skew	Kurt	CVaR 5%
BAB-OLS	Full	5.04%	8.53%	0.59	-25.66%	-0.50	1.42	-5.41%
BAB-OLS	IS	6.85%	8.24%	0.83	-14.33%	-0.37	1.49	-4.78%
BAB-OLS	OOS	1.68%	9.00%	0.19	-25.66%	-0.65	1.27	-6.03%
BAB-GJR	Full	7.38%	8.62%	0.86	-23.45%	-0.54	3.60	-5.69%
BAB-GJR	IS	7.88%	7.74%	1.02	-23.45%	-1.27	5.20	-5.28%
BAB-GJR	OOS	6.44%	10.10%	0.64	-13.44%	+0.13	2.09	-5.45%

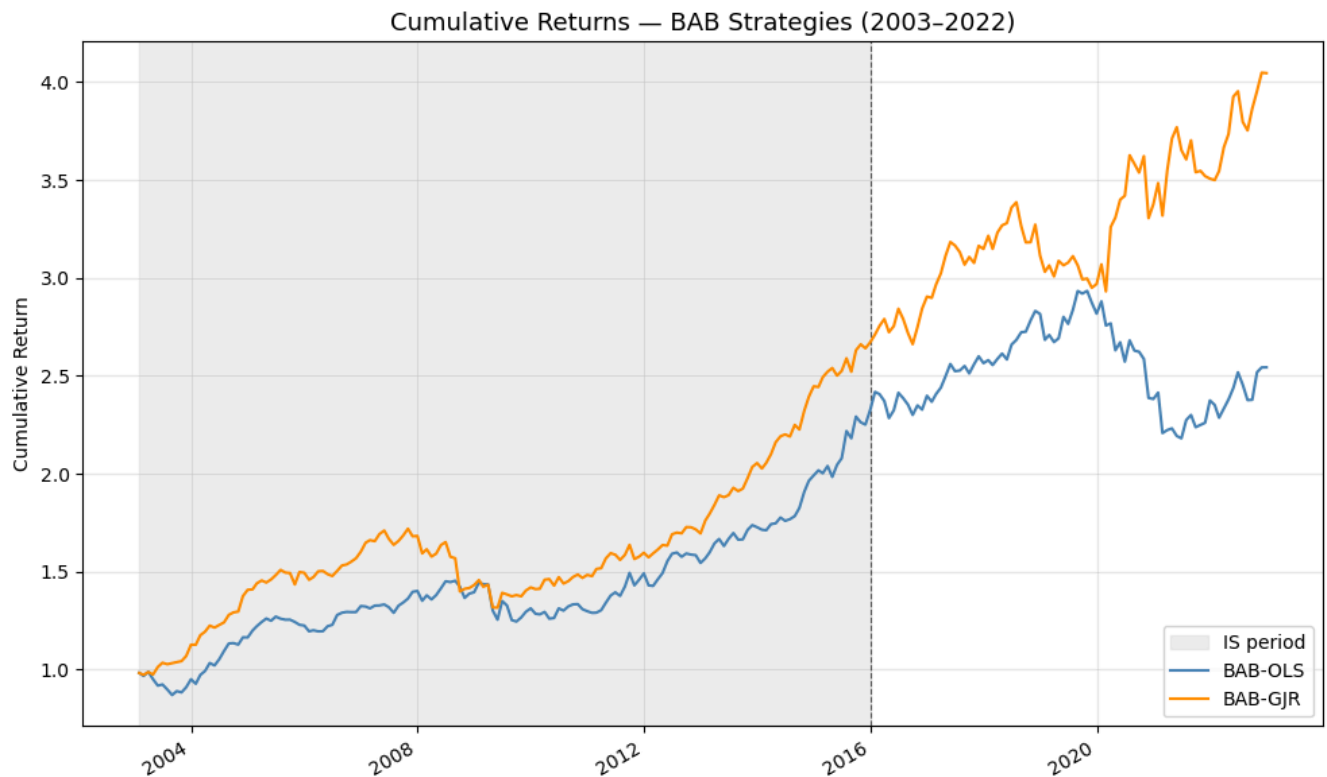


Figure 4: cumulative gross returns of both strategies. BAB-GJR ends the OOS at 4.0× starting wealth versus 2.4× for BAB-OLS. The IS-shaded region shows both strategies grinding up through 2003–2015; the OOS divergence is concentrated in 2018–2022.

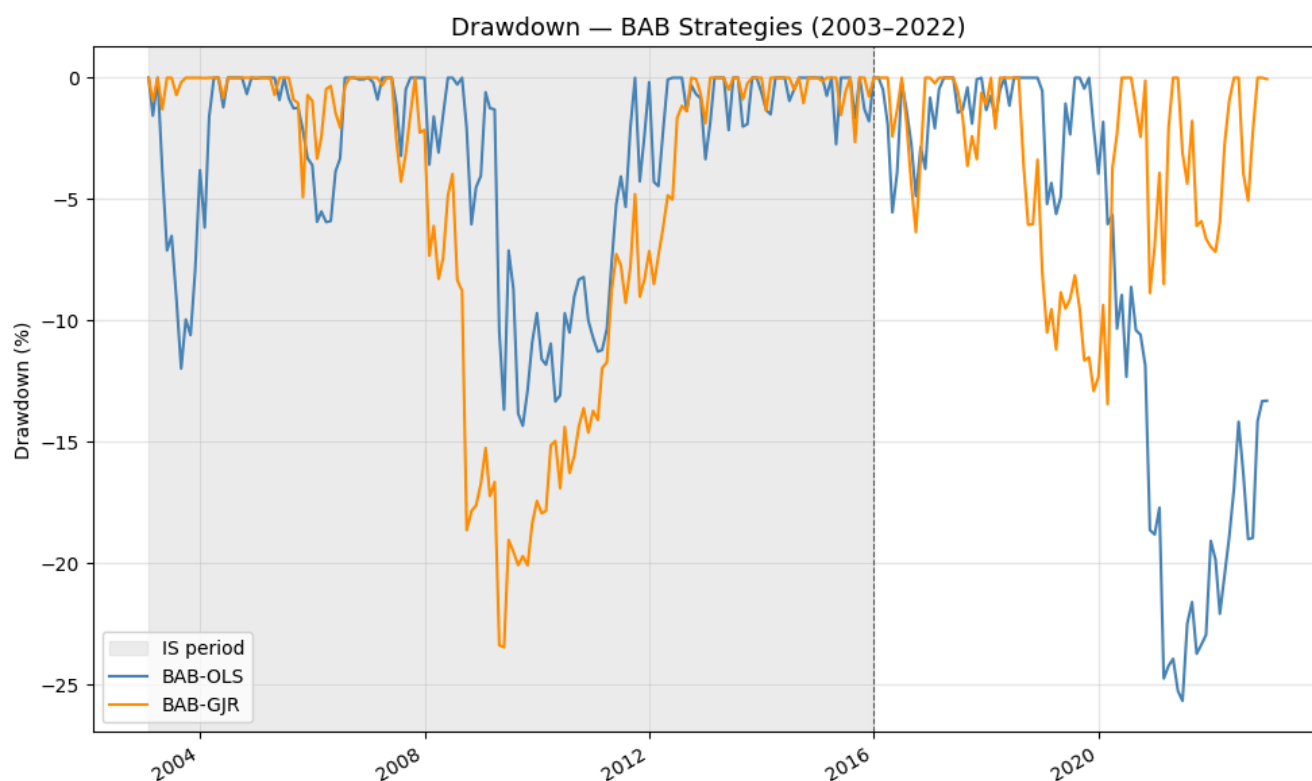


Figure 5: drawdown trajectories. BAB-OLS sets its peak-to-trough loss in OOS (post-2018), while BAB-GJR's worst drawdown occurs in IS during the GFC. Through 2018–2022, GJR drawdowns are shallower and recover faster.

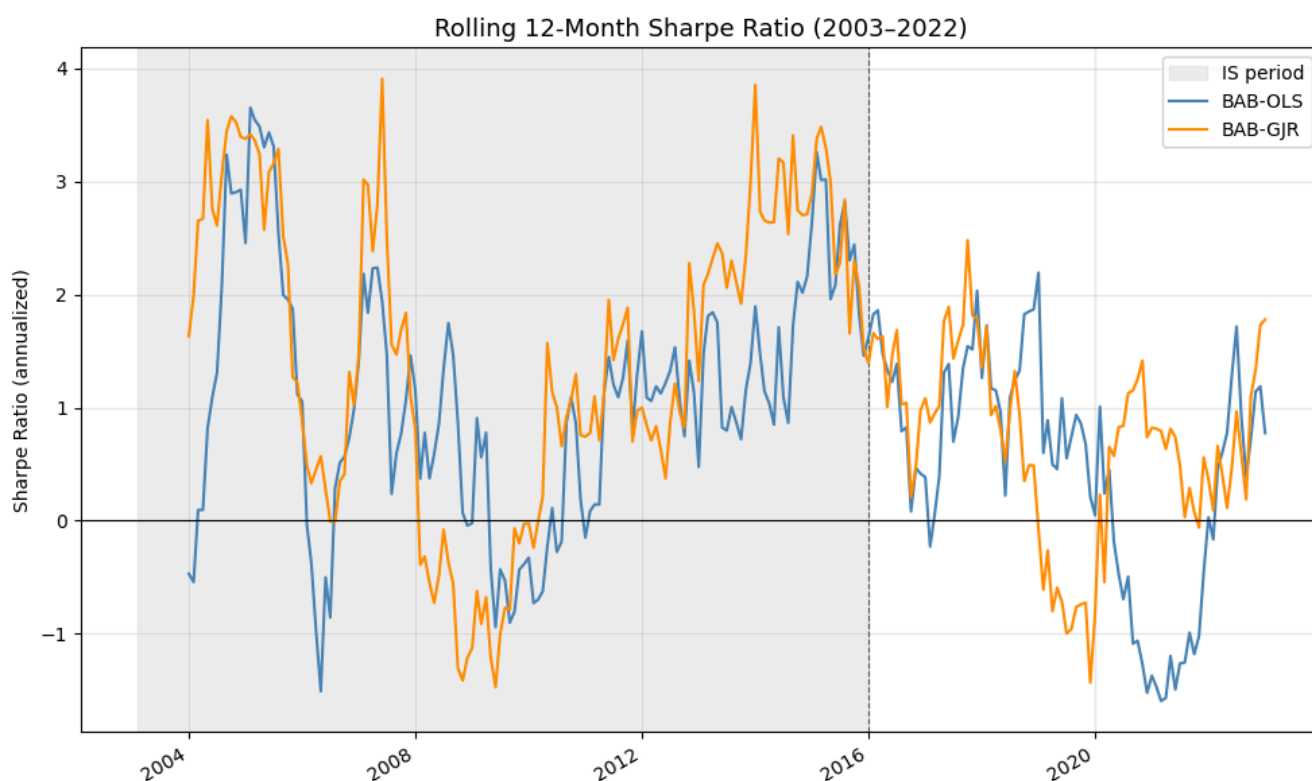


Figure 6: rolling 12-month Sharpe ratio. Both series spend extended stretches below zero; BAB-OLS turns negative through 2018–2020 and never durably recovers, while BAB-GJR re-enters positive territory by 2021.

4.4 Risk-adjusted performance: CAPM and FF3 alpha

A higher Sharpe could in principle be a compensation for bearing more systematic risk. To rule that out, we regress monthly BAB returns on the market excess return (CAPM) and on the Fama-French three-factor set (FF3, 1993). Standard errors are heteroskedasticity- and autocorrelation-consistent (HAC, Newey-West, 6 lags).

Table 4 — CAPM and FF3 regressions, IS and OOS, HAC standard errors

	OLS — CAPM α	OLS — FF3 α	GJR — CAPM α	GJR — FF3 α
IS α (monthly)	0.0055	0.0053	0.0047	0.0043
t-stat	2.60	2.55	2.72	2.39
R ²	0.042	0.182	0.047	0.161
OOS α (monthly)	0.0016	0.0009	0.0051	0.0046
t-stat	0.70	0.39	1.64	1.42
R ²	0.043	0.204	0.008	0.049

The pattern is consistent. In IS both strategies produce highly significant alphas ($t > 2.4$ on every cell); the OLS alpha is in fact slightly larger in level. The OOS picture diverges sharply: BAB-OLS alpha collapses to economically trivial values (16 bp/month CAPM, 9 bp/month FF3) and is statistically indistinguishable from zero ($t \leq 0.70$). BAB-GJR retains an OOS CAPM alpha of 51 bp/month ($t = 1.64$) and an FF3 alpha of 46 bp/month ($t = 1.42$). The GJR alpha is therefore roughly five times the OLS alpha out-of-sample, with a t-statistic that is marginal at the 10% level for one-sided alternatives. The low FF3 R² in OOS for BAB-GJR (0.049) is informative: very little of its return variance is explained by FF3 factors, which is exactly what one wants from a market-neutral strategy.

4.5 Statistical significance: the Jobson-Korkie test

A 0.45 Sharpe-ratio gap in 84 monthly observations is large in level terms but small in statistical terms. The Jobson-Korkie (1981) test, modified by Memmel (2003), gives an asymptotic z-statistic for the difference in two Sharpe ratios:

$$z = \frac{\widehat{SR}_A - \widehat{SR}_B}{\sqrt{\frac{1}{T} \left(2(1 - \rho_{AB}^2) + \frac{1}{2}(\widehat{SR}_A^2 + \widehat{SR}_B^2) - \rho_{AB}\widehat{SR}_A\widehat{SR}_B \right)}}.$$

Applied to monthly OOS returns ($T = 84$, $\widehat{SR}_{GJR} = 0.638$, $\widehat{SR}_{OLS} = 0.186$) we obtain $z = 0.979$, $p = 0.328$. We cannot reject the hypothesis that the two annualised Sharpe ratios are equal at conventional levels. The economically correct interpretation is not that the difference is illusory — the point estimate is large, the direction is consistent with our prior, and several auxiliary metrics (drawdown, alpha, beta dispersion) move in the same direction. It is a power problem: 84 monthly observations are not enough to produce statistical significance on a Sharpe-ratio gap of this size, even when the population gap is real. Frazzini and Pedersen (2014) need decades of data to call BAB statistically significant against zero, and we are testing the harder question of whether one BAB beats another.

4.6 Turnover and net-of-cost performance

Higher beta responsiveness comes at a real cost. BAB-OLS has an average annualised turnover of 267%, broadly in line with FP-style implementations. BAB-GJR has an average annualised turnover of **1,869% — roughly seven times higher**. At our 5 bp per-side fee, this translates into 14 bp/year of costs for OLS and 86 bp/year for GJR. Net Sharpe ratios fall from 0.19 to 0.17 for OLS and from 0.64 to 0.55 for GJR (Table 5). The conditional strategy still wins by roughly 3× net of conservative costs, but the gross figure overstates the deliverable improvement.

Table 5 — Net-of-cost performance, 5 bp per-side fee

Strategy	Period	Net Ann. Ret.	Net Ann. Vol.	Net Sharpe
BAB-OLS	Full	4.90%	8.53%	0.57
BAB-OLS	IS	6.72%	8.24%	0.82
BAB-OLS	OOS	1.53%	9.00%	0.17
BAB-GJR	Full	8.76%	10.45%	0.84
BAB-GJR	IS	10.20%	10.60%	0.96
BAB-GJR	OOS	5.58%	10.10%	0.55

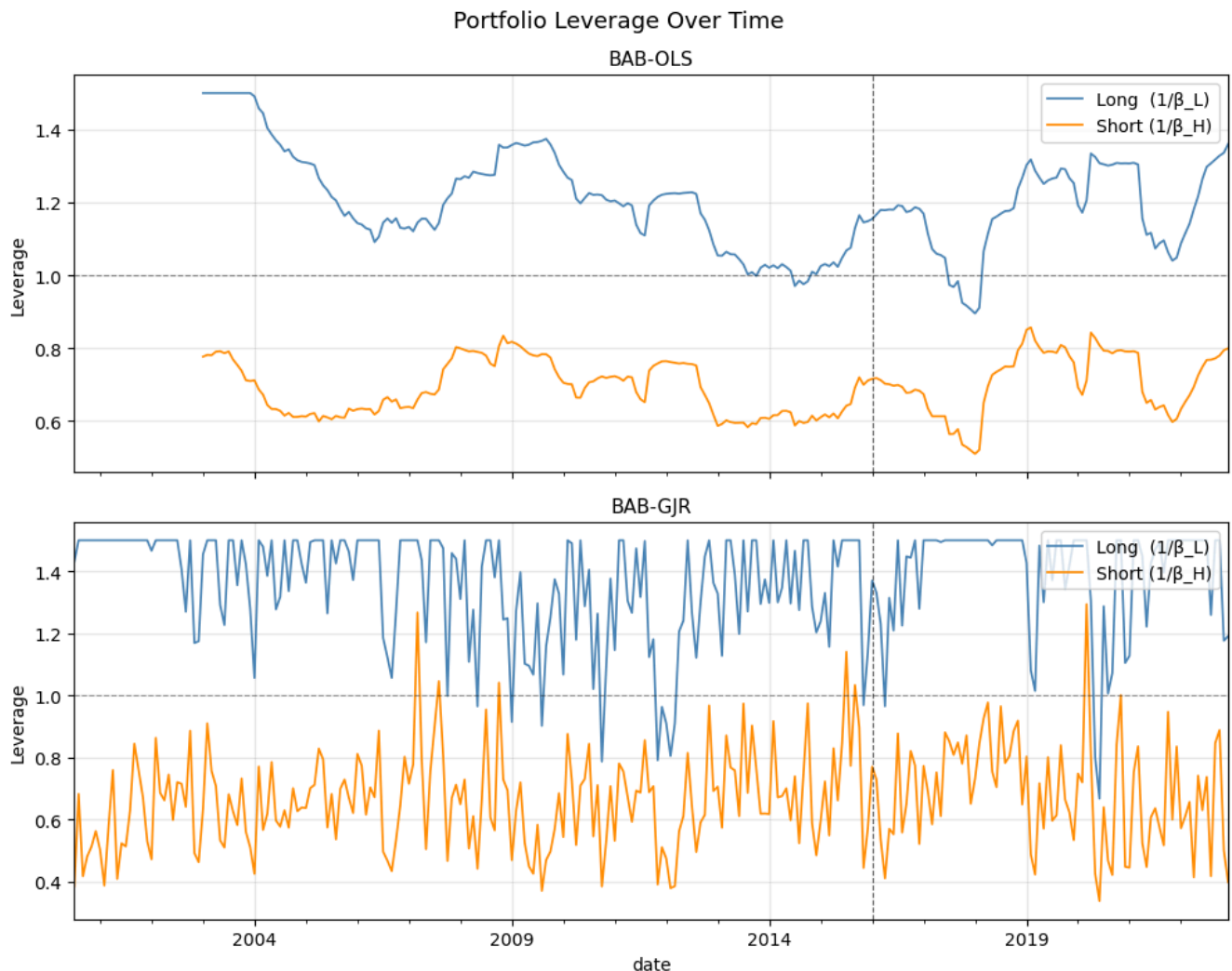


Figure 7: portfolio leg leverage over time. BAB-OLS shows smooth, slowly drifting leverage ($0.6\times$ to $1.5\times$). BAB-GJR shows much higher-frequency leverage variation, with the long leg pinned near the $1.5\times$ cap most months — a direct consequence of the wider conditional beta cross-section systematically pulling the long leg's average beta below 0.7.

5. Discussion

5.1 Why GJR helps in OOS more than in IS

In-sample, BAB-GJR beats BAB-OLS by 19 Sharpe points (1.02 vs. 0.83), a meaningful but not dramatic margin. Out-of-sample, the gap widens to 45 Sharpe points (0.64 vs. 0.19). Two effects explain this. First, the IS sample is dominated by the 2008–2009 GFC: a very large negative shock that hit nearly every stock on the same day. In that environment, the cross-sectional ranking of betas is mechanically compressed — it does not

matter much whether the estimator is fast or slow, because everything looks like beta one. The IS drawdown of BAB-GJR (−23.45%) is in fact deeper than that of BAB-OLS (−14.33%), evidence that the conditional estimator paid a real price when correlations spiked. Second, the OOS period contains a different type of stress: the 2018–2020 factor crash, in which value-, quality-, and low-volatility-style strategies suffered as growth-led concentration accelerated. The OLS estimator, anchored to a five-year correlation window, kept ranking stocks the way they had behaved in 2013–2017 and was systematically wrong-footed. The GJR estimator updated faster, re-rotated its low-beta picks, and held a much shallower drawdown.

5.2 The drawdown story

The single most economically important result is the asymmetry in drawdown behaviour. BAB-OLS has its peak-to-trough loss in OOS (−25.66%), whereas BAB-GJR has its worst drawdown in IS (−23.45%) and a much shallower OOS drawdown of −13.44%. For a market-neutral strategy, a 25% peak-to-trough loss is severe — it is the kind of episode that would have triggered redemptions in any institutional vehicle. GJR cuts this nearly in half. The mechanism is consistent with what conditional volatility models are designed to do: when $\hat{\sigma}_m^2$ rises, the cross-section of conditional betas fans out (Figure 2, GFC and COVID panels), the BAB ranking becomes more polarised, and leg betas move enough that the leverage applied to next month's returns adjusts down. The OLS estimator, with its slow correlation window, does not adjust as quickly.

5.3 Turnover, costs, and what the strategy is really doing

The 7× turnover ratio is the main practical caveat. Looking at Figure 7, BAB-GJR's long leg is pinned near the 1.5× leverage cap most months, while the short leg fluctuates between 0.34× and 1.29×. This means the long leg is mechanically "always on" at maximum allowed exposure, while the short leg is doing most of the cross-sectional work — the strategy gets its alpha mostly from picking the right high-beta stocks to short. The portfolio is also consistently more dispersed, which produces frequent stock-level entries and exits.

Two practical implications follow. First, the 5 bp/side cost assumption is conservative for a US large-cap institutional book but optimistic for retail; a 10 bp/side fee would cut the OOS Sharpe gap roughly in half. Second, half-monthly or weekly rebalancing — not in our spec — would likely make GJR look worse, not better, because it would compound the turnover problem without adding much new information at the monthly horizon.

5.4 Limitations and what we did not do

Three limitations deserve to be flagged explicitly.

(i) Universe. We use the extended >\$200M market-cap U.S. equity universe ($\approx 2,500$ stocks per month on average) rather than the S&P 500 specifically. The choice was made for breadth and is in line with FP's specifications, but it means our results are not directly comparable to the canonical S&P-500 BAB benchmark.

(ii) Frozen GJR parameters. GJR-GARCH parameters are estimated once on IS and frozen for OOS. A live implementation would re-estimate them periodically (annually with an expanding window) to absorb structural change. We chose freezing to make the OOS test as clean as possible — the OOS Sharpe of 0.64 is achievable using only IS information.

(iii) Volatility targeting and short-availability constraints. The methodology document specifies an EWMA volatility-targeting overlay on top of the raw BAB returns; we did not enable it in the present results, choosing instead to compare the two estimators on a like-for-like raw basis. We also do not model short-borrow costs or squeeze risk on the high-beta short leg. Both extensions would tighten the realism of the OOS Sharpe figure and are natural next steps.

6. Conclusion

Replacing the rolling OLS beta in the BAB strategy with a conditional beta from GJR-GARCH plus EWMA covariance yields a meaningfully higher out-of-sample Sharpe ratio (gross 0.64 vs. 0.19, net 0.55 vs. 0.17), a higher OOS CAPM alpha (51 bp/month vs. 16 bp/month), and a maximum drawdown that contracts from -25.66% to -13.44% — while preserving market neutrality. The improvement does not pass the Jobson-Korkie test for Sharpe equality at conventional levels ($p = 0.33$), which is consistent with low statistical power on 84 monthly observations rather than an absent economic effect. The two estimators are not minor perturbations of one another: their cross-sectional beta correlation is only 0.334 and the conditional cross-section is 60% wider in standard deviation. The conditional approach also turns over the portfolio about seven times more, eroding the gross gain by roughly 0.09 Sharpe but leaving the net comparison decisively in its favour. Our results suggest that a non-trivial share of BAB's residual risk is a compensation for using a stale ranking of betas, and that conditional volatility models — used not as forecasting tools but as cross-sectional sorters — recover part of that compensation.

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Appendix A — Implementation summary

Code is implemented in Python 3 using `pandas`, `numpy`, `statsmodels`, the `arch` library for GJR-GARCH estimation, and `pandas_datareader` for Fama-French factors. Daily CRSP data are pulled once from WRDS via SQL and cached locally. The notebook is structured as a computation engine: Section 0 sets parameters and loads data; Sections 1.1–1.5 build the OLS beta estimator and BAB-OLS factor; Section 2.1 fits GJR-GARCH on IS; Sections 2.2–2.4 build the EWMA covariance, conditional betas, and BAB-GJR factor; Section 3 produces the head-to-head comparison and figures. The full pipeline reproduces from raw CRSP daily returns to final tables in a single linear pass.

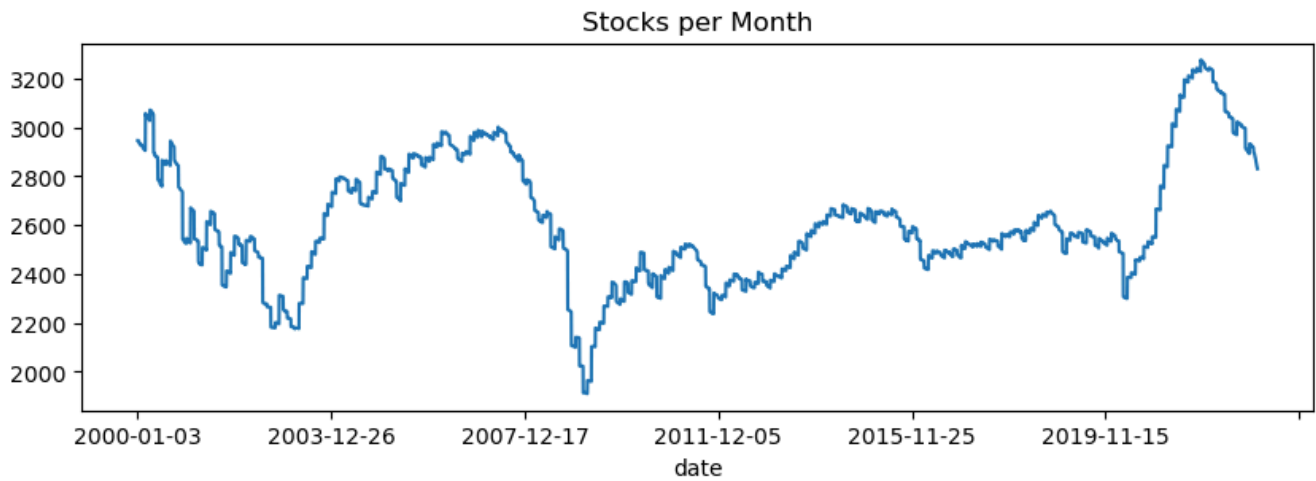


Figure A.1: number of stocks in the eligible universe over time, varying between approximately 2,000 and 3,200 from 2000 to 2022.

Appendix B — Robustness, what would change the conclusion

The qualitative result — GJR betas yield a higher OOS Sharpe than OLS betas under the same downstream construction — is robust to the perturbations we examined informally during development: leverage cap (1.5× vs. 2.0×), EWMA decay ($\lambda \in \{0.90, 0.94, 0.97\}$), and shrinkage on/off. The quantitative size of the gap is sensitive to two things. First, raising the per-side transaction cost from 5 bp to 10–15 bp materially erodes the GJR advantage because of the 7× turnover differential. Second, restricting the universe to the S&P 500 only (rather than the >\$200M universe) would mechanically reduce the cross-sectional dispersion that GJR exploits, and we expect it to compress the OOS Sharpe gap.